Marmara University
Electrical and Electronics Engineering Department
Fall 2023
EE 3016/3051 – Electromagnetics I
Simulation Project - Due December 17, 2023

Numerical Solution of Poisson's Equation

The main purpose of this simulation project is to numerically solve a boundary-value problem involving the electric potential. The electric potential distribution inside a given structures is obtained numerically. The finite-difference (FD) method is used to solve Poisson's equation in two dimensions

Introduction

The electric potential distribution inside any given structure can be analytically obtained by solving Poisson's equation subject to some boundary conditions. If we assume no volume charge inside the structure, Poisson's equation becomes (Laplace's equation)

$$\nabla^2 V = 0 \tag{1}$$

In two dimensions (rectangular coordinates), equation (1) is written as

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial x^2} = 0 \tag{2}$$

The ability to solve equation (2) depends in a great deal on the nature of the structure under consideration. In some cases a closed-form (analytical) solution to equation (2) is difficult to obtain. Alternatively, numerical methods can be used, especially for large problems. Numerical methods utilize the speed of computers and the flexibility of programming.

In this project, the Finite-Difference (FD) method will be used. The FD method is one of the most popular numerical methods in the field of electromagnetics. Its main advantages are (1) easy to formulate, (2) suitable for many structures and (3) flexible for modifications.

The FD-based solution is performed in the following steps.

1) Discretize the given structure (make a grid) using a suitable step size in both dimensions (Δx and Δy). The accuracy of the results improves with smaller step sizes.
2) Approximate the partial derivatives in equation (2) by the following second-order relations.

$$\frac{\partial^2 V}{\partial x^2} = \frac{V(i+1,j) - 2V(i,j) + V(i-1,j)}{\Delta x^2} \tag{3}$$

$$\frac{\partial^2 V}{\partial y^2} = \frac{V(i,j+1) - 2V(i,j) + V(i,j-1)}{\Delta y^2} \tag{4}$$

   where i and j are the indices along the x- and the y-axis, respectively. Substituting equations (3) and (4) in equation (2), and using $\Delta x = \Delta y$, we get

$$V(i,j) = \frac{1}{4}[V(i+1,j) + V(i-1,j) + V(i,j+1) + V(i,j-1)] \tag{5}$$

3) Solve the resulting linear system of equations for the unknown potential nodes.

Example:

Find the potential distribution inside the structure given in figure 1. Take a step size of 5 cm in both dimensions. The boundary conditions are as shown.

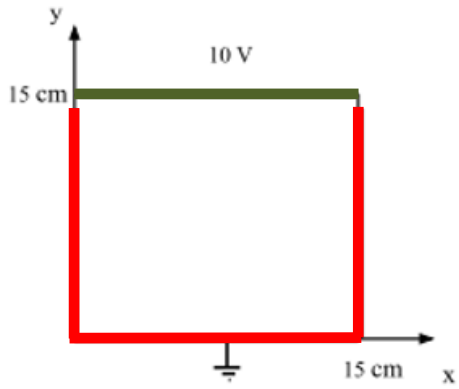


Fig. 1. The structure of the example.

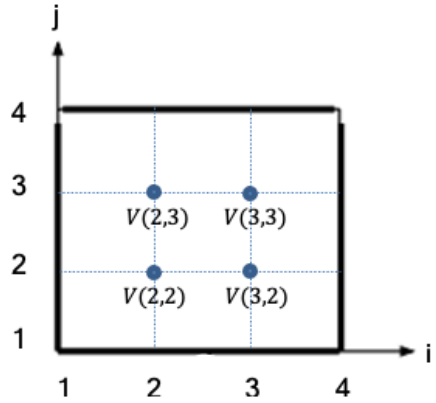


Fig. 2. The grid used for solution.

The grid is arranged as shown in figure 2. At each node in the figure, the value of the potential is labeled. Only four unknown values are to be determined; the potential value for all other nodes are given as boundary conditions (for example, $V(1,1)=0$, $V(4,3)=0$, $V(4,4)=10$).

Applying the algorithm of equation (5), we get the following system of linear equations.

$$4V(2,2) = V(3,2) + 0 + V(2,3) + 0$$
$$4V(2,3) = V(3,3) + 0 + 10 + V(2,2)$$
$$4V(3,2) = 0 + V(2,2) + V(3,3) + 0$$
$$4V(3,3) = 0 + V(2,3) + 10 + V(3,2)$$

The above equations can be arranged in a distinct diagonal matrix form, as follows.

$$\begin{pmatrix} 4 & -1 & -1 & 0 \\ -1 & 4 & 0 & -1 \\ -1 & 0 & 4 & -1 \\ 0 & -1 & -1 & 4 \end{pmatrix} \begin{pmatrix} V(2,2) \\ V(2,3) \\ V(3,2) \\ V(3,3) \end{pmatrix} = \begin{pmatrix} 0 \\ 10 \\ 0 \\ 10 \end{pmatrix}$$

The solution of the above system, which is in the form $Ax = B$, can be obtained using

$$x = A^{-1}B$$

This operation can be easily implemented in MATLAB. The result is:

$V(2,2)=1.25$ V, $V(2,3)=3.75$ V, $V(3,2)=1.25$ V, $V(3,3)=3.75$ V

Required Work

1. Solve Laplace's equation using the FD method for the structure given below (figure 3). The boundary of the left half of the structure is biased at -1V, whereas the boundary of the right half is biased at +5V. Use a step size of 2 cm in both dimensions.
2. Decrease the step size to 1 cm, 0.5 cm and 0.25 cm and repeat part (1).
3. Compare the results obtained in parts (1) and (2) for some points inside the structure. Make error analysis (a plot would be a good idea).
4. Produce contour plots for the equi-potential lines inside the structure. (In MATLAB, there is a function for contouring.)
5. Produce a vector plot of the electric field inside the structure (bonus marks).

6. Make one modification to the given structure (unique to your team) by introducing some features inside the basic structure (see examples below). Find the new potential distribution contour. Give comments.

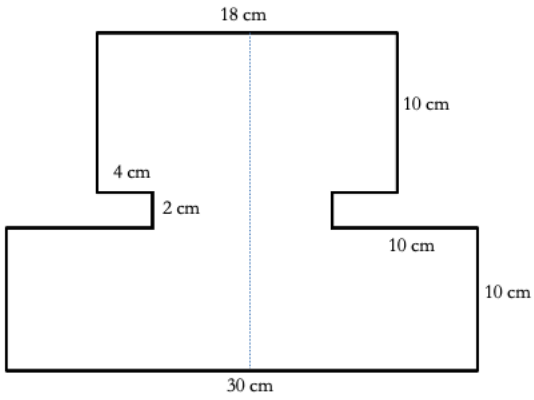
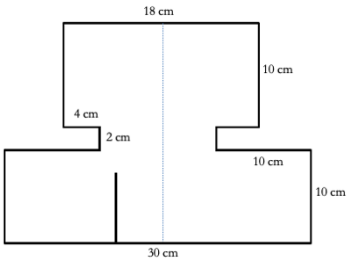
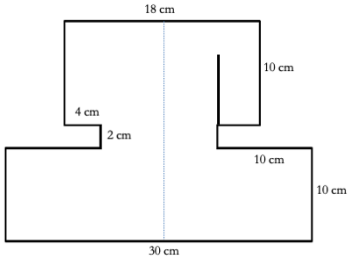


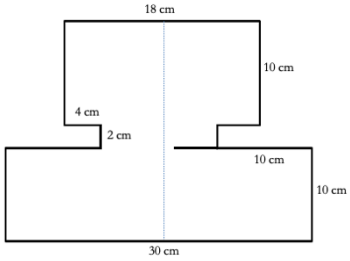
Fig. 3. The structure to be analyzed.

Instructions

- Use MATLAB for solution and plotting.
- Up to 4 students can work together as a team.
- Submissions should be typed and in a formal report format.
- Prepare your project report (one file) in pdf format.
- In case of copied work (analysis, graphs, etc., even partially), all involved parties will get nulls.







Upload link: https://www.dropbox.com/request/x3BB4Hoq1aZJJbTYs6gd

---

Show all work and analysis
Only pdf files are acceptable
Late submissions are checked but not graded
Copied submissions represent academic dishonesty and are disqualified